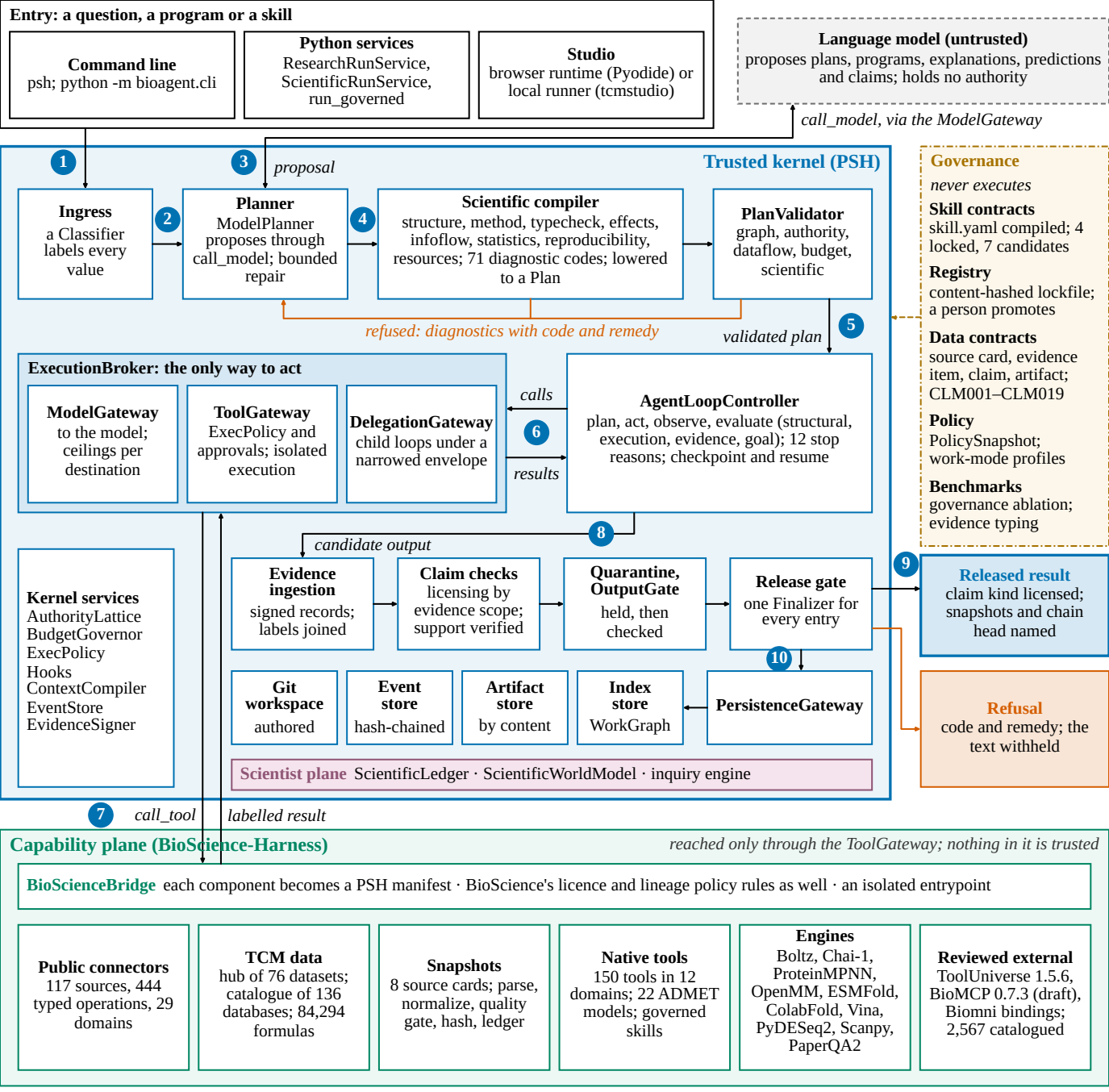
Entry: a question, a program or a skill
Command line
psh; python -m bioagent.cli
Python services
ResearchRunService, ScientificRunService, run_governed
Studio
browser runtime (Pyodide) or local runner (tcmstudio)
Language model (untrusted)
proposes plans, programs, explanations, predictions and claims; holds no authority
call_model, via the ModelGateway
Trusted kernel (PSH)
1
Ingress
a Classifier labels every value
2
Planner
ModelPlanner proposes through call_model; bounded repair
3
proposal
4
Scientific compiler
structure, method, typecheck, effects, infoflow, statistics, reproducibility, resources; 71 diagnostic codes; lowered to a Plan
PlanValidator
graph, authority, dataflow, budget, scientific
refused: diagnostics with code and remedy
validated plan
5
ExecutionBroker: the only way to act
ModelGateway
to the model; ceilings per destination
ToolGateway
ExecPolicy and approvals; isolated execution
DelegationGateway
child loops under a narrowed envelope
calls
6
results
AgentLoopController
plan, act, observe, evaluate (structural, execution, evidence, goal); 12 stop reasons; checkpoint and resume
8
candidate output
Kernel services
AuthorityLattice
BudgetGovernor
ExecPolicy
Hooks
ContextCompiler
EventStore
EvidenceSigner
Evidence ingestion
signed records; labels joined
Claim checks
licensing by evidence scope; support verified
Quarantine, OutputGate
held, then checked
Release gate
one Finalizer for every entry
9
Released result
claim kind licensed; snapshots and chain head named
10
Git workspace
authored
Event store
hash-chained
Artifact store
by content
Index store
WorkGraph
PersistenceGateway
Refusal
code and remedy; the text withheld
Scientist plane ScientificLedger · ScientificWorldModel · inquiry engine
Governance
never executes
Skill contracts
skill.yaml compiled; 4 locked, 7 candidates
Registry
content-hashed lockfile; a person promotes
Data contracts
source card, evidence item, claim, artifact; CLM001–CLM019
Policy
PolicySnapshot; work-mode profiles
Benchmarks
governance ablation; evidence typing
7
call_tool
labelled result
Capability plane (BioScience-Harness)
reached only through the ToolGateway; nothing in it is trusted
BioScienceBridge each component becomes a PSH manifest · BioScience's licence and lineage policy rules as well · an isolated entrypoint
Public connectors
117 sources, 444 typed operations, 29 domains
TCM data
hub of 76 datasets; catalogue of 136 databases; 84,294 formulas
Snapshots
8 source cards; parse, normalize, quality gate, hash, ledger
Native tools
150 tools in 12 domains; 22 ADMET models; governed skills
Engines
Boltz, Chai-1, ProteinMPNN, OpenMM, ESMFold, ColabFold, Vina, PyDESeq2, Scanpy, PaperQA2
Reviewed external
ToolUniverse 1.5.6, BioMCP 0.7.3 (draft), Biomni bindings; 2,567 catalogued